

TECHNICAL TRAINING

Python Programming & Project Development

100 Days of Code™: The Complete Python Pro Bootcamp

Python • PyCharm • Project-Based Learning

STUDENT CREDENTIALS

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ROLL NO.
25SCS1003004835

SECTION
2CSE4

SESSION
2025–2029

Training Overview

Self-paced, project-based Python learning

ABOUT THE COURSE

100 Days of Code™: The Complete Python Pro Bootcamp by Dr. Angela Yu.

The course uses practical programming exercises and mini projects to build Python skills from fundamentals onward.

KEY AREAS

- Python fundamentals
- Control flow
- Functions
- Strings and lists
- Dictionaries
- Randomization
- User input
- Debugging
- Console applications

Projects Developed

Five practical Python applications

- | | | |
|----|----------------------------|---------------------------------------|
| 01 | Password Generator | Randomized password creation |
| 02 | Rock Paper Scissors | Interactive game against the computer |
| 03 | Caesar Cipher | Text encryption and decryption |
| 04 | Hangman Game | Word guessing with lives |
| 05 | Blind Auction | Bidding and highest-bid selection |

01 Password Generator

Randomized password creation

PYCHARM — CODE

```
1 import random
2 letters = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r',
3           's', 't', 'u', 'v', 'w', 'x', 'y', 'z']
4
5 numbers = ['1', '2', '3', '4', '5', '6', '7', '8', '9', '0']
6
7 symbols = ['!', '@', '$', '%', '^', '&', '*', '(', ')', '+']
8
9 print("welcome to the pypassword generator")
10
11 nr_letter = int(input("enter the number of letters: "))
12 nr_symbol = int(input("enter the number of symbols: "))
13 nr_number = int(input("enter the number of numbers: "))
14
15 password_list = []
16
17 for char in range(0, nr_letter):
18     password_list.append(random.choice(letters))
19 for char in range(0, nr_symbol):
20     password_list.append(random.choice(symbols))
21 for char in range(0, nr_number):
22     password_list.append(random.choice(numbers))
23
24 print(password_list)
25 random.shuffle(password_list)
26 print(password_list)
27
28 password = ""
29 for char in password_list:
30     password += char
31 print(f"your password is: {password}")
32
```

TERMINAL — OUTPUT

```
aashi@Awan-Laptop-Personal: /mnt/c/Users/Dell/OneDrive/Documents/coding/python$ python password.py
welcome to the pypassword generator
enter the number of letters: 2
enter the number of symbols: 3
enter the number of numbers: 4
['b', 'u', '8', '$', 'x', '9', '3', '0', '5']
['u', '8', '0', '$', '0', '5', '3', 'x', '9']
your password is: u80$530
```

OBJECTIVE Generate a password from letters, symbols and numbers, then shuffle the characters.

01 Password Generator — Output

Example execution

PYCHARM — CODE

```
1 import random
2 letters = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r',
3           's', 't', 'u', 'v', 'w', 'x', 'y', 'z']
4
5 numbers = ['1', '2', '3', '4', '5', '6', '7', '8', '9', '0']
6
7 symbols = ['!', '#', '$', '%', '^', '&', '*', '(', ')', '+']
8
9 print("welcome to the pypassword generator")
10
11 nr_letter = int(input("enter the number of letters: "))
12 nr_symbol = int(input("enter the number of symbols: "))
13 nr_number = int(input("enter the number of numbers: "))
14
15 password_list = []
16
17 for char in range(0, nr_letter):
18     password_list.append(random.choice(letters))
19 for char in range(0, nr_symbol):
20     password_list.append(random.choice(symbols))
21 for char in range(0, nr_number):
22     password_list.append(random.choice(numbers))
23
24 print(password_list)
25 random.shuffle(password_list)
26 print(password_list)
27
28 password = ""
29 for char in password_list:
30     password += char
31 print(f"your password is: {password}")
32
```

TERMINAL — OUTPUT

```
aashi@Awan-Laptop-Personal: /mnt/c/Users/Dell/OneDrive/Documents/coding/python$ python password.py
welcome to the pypassword generator
enter the number of letters: 2
enter the number of symbols: 3
enter the number of numbers: 4
['b', 'u', '8', '$', '%', '9', '3', '0', '5']
['u', '8', '0', '$', '0', '5', '3', '2', '9']
your password is: u80$530
```

OBJECTIVE The program accepts the requested character counts and produces a randomized password.

02 Rock Paper Scissors

Player-versus-computer game

PYCHARM — CODE (PART 1)

```
python > rockpaperscissor > ...
1
2 import random
3
4 rock = """
5
6 ---'
7      ( )
8      ( )
9      ( )
10 ---'
11 """
12
13 paper = """
14
15 ---'
16      ( )
17      ( )
18      ( )
19 ---'
20 """
21
22 scissors = """
23
24 ---'
25      ( )
26      ( )
27      ( )
28 ---'
29 """
```

PYCHARM — CODE (PART 2)

```
30 game_images = [rock, paper, scissors]
31 user_choice = int(input("what do you choose, type 0 for rock, type 1 for paper, type 2 for scissors: "))
32 if user_choice >= 0 and user_choice <= 2:
33     print(game_images[user_choice])
34     computer_choice = random.randint(0,2)
35     print("computer chose:")
36     print(game_images[computer_choice])
37 if user_choice >= 3 or user_choice < 0:
38     print("you chose an invalid number")
39 elif computer_choice == 0 and user_choice == 2:
40     print("you lose")
41 elif user_choice == 0 and computer_choice == 2:
42     print("you win")
43 elif user_choice > computer_choice:
44     print("you win")
45 elif computer_choice > user_choice:
46     print("you lose")
47 elif computer_choice == user_choice:
48     print("its a draw")
```

OBJECTIVE Use random computer choices and conditional logic to determine the result.

02 Rock Paper Scissors — Output

Example execution

PYCHARM — CODE

```
python > rockpaperscissor > ...
1
2 import random
3
4 rock = """
5
6 ---'
7
8
9
10 ---'
11 """
12
13 paper = """
14
15 ---'
16
17
18
19
20 ---'
21 """
22
23 scissors = """
24
25 ---'
26
27
28 ---'
29 """
```

TERMINAL — OUTPUT

```
asahid@Pawan-Laptop-Personal: /mnt/c/Users/Dell/OneDrive/Documents/coding/python$ python rockpaperscissor
what do you choose, type 0 for rock, type 1 for paper, type 2 for scissors: 2

---'
---'
computer chose:

---'
---'
you win
```

OBJECTIVE The run shows the player's choice, the computer's choice and the final game result.

03 Caesar Cipher

Text encryption and decryption

PYCHARM — CODE

```
8 print(logo)
9 alphabet = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l',
10            'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z']
11
12
13
14 def caesar(original_text, shift_amount, encode_or_decode):
15     output_text = ""
16     if encode_or_decode == "decode":
17         shift_amount *= -1
18     for letter in original_text:
19         if letter not in alphabet:
20             output_text += letter
21         else:
22             shifted_position = alphabet.index(letter) + shift_amount
23             shifted_position %= len(alphabet)
24             output_text += alphabet[shifted_position]
25     print(f"here is the {encode_or_decode}d text: {output_text}")
26
27
28 should_continue = True
29 while should_continue:
30     direction = input("type 'encode' to encode text or 'decode' to decode text: ")
31     text = input("type your text: ").lower()
32     shift = int(input("type the shift number: "))
33
34     caesar(original_text=text, shift_amount=shift, encode_or_decode=direction)
35     restart = input("type 'yes' to continue or 'no' to exit:\n ").lower()
36
37 if restart == "no":
38     should_continue = False
39     print("Thank you for using caesar!")
```

TERMINAL — OUTPUT

```
aashi@Pawan-Laptop-Personal:/mnt/c/Users/Dell/OneDrive/Documents/coding/python$ python caesar.py
,adPPYba, ,adPPYba, ,adPPYba, ,adPPYba, ,adPPYba,
aB' "" "Y8 aB' "8 aB' "8 aB' "8
8b ,adPPPP88 8b----- 8b----- 8b-----
"8a, ,aa 88, ,88 "8b, ,aa "8b, ,aa
"Ybbd8" "8b8dP"Y8 "Ybbd8" "Ybbd8"
type 'encode' to encode text or 'decode' to decode text: encode
type your text: hello
type the shift number: 4
here is the encoded text: linnx
```

OBJECTIVE Shift alphabetic characters by a selected amount and reverse the shift when decoding.

03 Caesar Cipher — Output

Example: hello with shift 4

PYCHARM — CODE

```

8 print(logo)
9 alphabet = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l',
10             'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z']
11
12
13
14 def caesar(original_text, shift_amount, encode_or_decode):
15     output_text = ""
16     if encode_or_decode == "decode":
17         shift_amount *= -1
18     for letter in original_text:
19         if letter not in alphabet:
20             output_text += letter
21         else:
22             shifted_position = alphabet.index(letter) + shift_amount
23             shifted_position %= len(alphabet)
24             output_text += alphabet[shifted_position]
25     print(f"here is the {encode_or_decode}d text: {output_text}")
26
27
28 should_continue = True
29 while should_continue:
30     direction = input("type 'encode' to encode text or 'decode' to decode text: ")
31     text = input("type your text: ").lower()
32     shift = int(input("type the shift number: "))
33
34     caesar(original_text=text, shift_amount=shift, encode_or_decode=direction)
35     restart = input("type 'yes' to continue or 'no' to exit:\n ").lower()
36
37     if restart == "no":
38         should_continue = False
39     print("Thank you for using caesar!")

```

TERMINAL — OUTPUT

[illegible]

OBJECTIVE Example result shown in the terminal: hello → lipps.

04 Hangman Game

Word guessing with lives and game state

PYCHARM — CODE (PART 1)

```
75 word_list = ["aardvark", "baboon", "camel"]
76
77 lives = 6
78
79 chosen_word = random.choice(word_list)
80 print(chosen_word)
81 placeholder = ""
82 word_length = len(chosen_word)
83 for position in range(word_length):
84     placeholder += "_"
85 print(placeholder)
86
87 game_over = False
88 correct_letters = []
89
90 while not game_over:
91     guess = input("Guess a letter: ")
92     display = ""
93     for letter in chosen_word:
94         if letter == guess:
95             display += letter
96             correct_letters.append(letter)
97         elif letter in correct_letters:
98             display += letter
99         else:
100             display += "_"
101     print(display)
102     if guess not in chosen_word:
103         lives -= 1
104         if lives == 0:
105             game_over = True
106             print("You Lose!")
```

PYCHARM — CODE (PART 2)

```
if "_" not in display:
    game_over = True
    print("You win!")

print(stages[lives])
```

OBJECTIVE Select a random word, reveal correct letters, reduce lives and detect win/lose states.

04 Hangman Game — Execution

Multiple stages of the game

OUTPUT — EARLY GUESSES

Guess a letter: h

```
  +---+
  |   |
  o   |
  |   |
  +---+
```

Guess a letter: j

```
  +---+
  |   |
  o   |
  |   |
  +---+
```

Guess a letter: p

```
  +---+
  |   |
  o   |
  |   |
  +---+
```

OUTPUT — GAME PROGRESS

Guess a letter: k

```
  +---+
  |   |
  o   |
  |   |
  +---+
```

Guess a letter: b

b _ _

```
  +---+
  |   |
  o   |
  |   |
  +---+
```

Guess a letter: g

b _ _

```
  +---+
  |   |
  o   |
  |   |
  +---+
```

Guess a letter: j

OUTPUT — FINAL STATE

b _ _

```
  +---+
  |   |
  o   |
  |   |
  +---+
```

Guess a letter: j

b _ _

You Lose!

```
  +---+
  |   |
  o   |
  |   |
  +---+
```

05 Blind Auction

Bidding simulation using dictionaries

PYCHARM — CODE

```
14 print(logo)
15
16 def find_highest_bidder(bidding_dictionary):
17     winner = ""
18     highest_bid = 0
19     for bidder in bidding_dictionary:
20         bid_amount = bidding_dictionary[bidder]
21         if bid_amount > highest_bid:
22             highest_bid = bid_amount
23             winner = bidder
24
25     print(f"the winner is {winner} with the highest bid of {highest_bid}" )
26
27
28
29 bids = {}
30 continue_bidding = True
31 while continue_bidding:
32     name = input("Enter your name: ")
33     price = int(input("what is your bid? $"))
34     bids[name] = price
35     should_continue = input("are there any other bidders? type 'yes' or 'no':\n ").lower()
36     if should_continue == "no":
37         continue_bidding = False
38         find_highest_bidder(bids)
39     elif should_continue == "yes":
40         print("\n"*20)
41
```

TERMINAL — OUTPUT



```
Enter your name: ashwina
what is your bid? $200
are there any other bidders? type 'yes' or 'no':
yes
```

OBJECTIVE Store bidder names and amounts, compare bids and identify the highest bidder.

05 Blind Auction — Results

Bidding and winner selection

TERMINAL — BIDDING



```
Enter your name: ashwina
what is your bid? $200
are there any other bidders? type 'yes' or 'no':
yes
```

TERMINAL — FINAL RESULT

```
Enter your name: ananya
what is your bid? $300
are there any other bidders? type 'yes' or 'no':
no
the winner is ananya with the highest bid of 300
```

OBJECTIVE The final run identifies the bidder with the highest submitted amount.

Skills Acquired

What I practiced through the course and projects

PYTHON SKILLS

- Variables and data types
- Conditions and loops
- Functions
- Strings and lists
- Dictionaries
- Randomization
- Input handling
- Debugging

PROJECT SKILLS

- Breaking problems into steps
- Implementing program logic
- Testing different inputs
- Reading terminal output
- Fixing errors
- Building complete console programs

Course Completion Certificate

Dedicated space for your certificate



Certificate no: UC-cccd9b34-751a-4be4-a0bf-2084fbb9a283
Certificate url: ude.my/UC-cccd9b34-751a-4be4-a0bf-2084fbb9a283
Reference Number: 0004

CERTIFICATE OF COMPLETION

100 Days of Code™ : The Complete Python Pro Bootcamp

Instructors **Dr. Angela Yu, Developer and Lead Instructor**

Ashwina

Date **Aug. 29, 2026**

Length **57 total hours**

CONCLUSION

Learning by building.

SUMMARY

The training strengthened my Python fundamentals through hands-on project development. The five applications demonstrate practical use of programming logic, data structures, functions, randomization, input handling and debugging.

Thank You

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